expect the third-party chip design services market in India to grow at a CAGR of about 22.3 per cent, to reach nearly $830 million by 2012."

- Over 125 companies are estimated to be India-involved in various aspects of design. There are the big multinationals such as IBM, Intel, Texas Instruments. Then there are the big Indian IT players like Wipro, Infosys and TCS, who have set up engineering design service practices and an ever growing number of entrepreneurial firms. All this makes India a hot destination for chip design and creating more job opportunities. The industry is estimated to employ over 1,00,000 people.

What are the related paths that you can pursue?

Chip design career can have the following flavours:

- SOC integration engineer
- IP developer: analogue or digital
- DST design engineer: focusing on testing
- Verification engineer: verify design in simulations

There are two separate cadres:

- **Technical cadre:** typically everybody starts on the technical side as individual contributors, i.e. design engineer, senior design engineer and moving up to division head.
- Over time, senior technical people can also assume part of the role of business development – in terms of understanding customers' needs and suggesting solutions.
- **Project manager cadre:** post 6-8 years of experience, designers can move into the project manager role involving management of project, integration and liaison.

Why choose this career?

Rahul Hakhoo, Head, Computer Peripheral Divisions, ST Microelectronics: It was a strange quirk of fate. While I was completing my engineering, one of my professors suggested I do an internship with a chip manufacturing company called Semiconductor Complex Ltd. As students, we had heard about chip design being the 'next big thing' and though I was quite at sea about it, I decided to join the unit. The internship led to a full-time job and within the first six months my supervisor asked me to shift to the design division of the company. There was so much to learn there that I ended up spending another six years in the design unit!

Face-2-Face with practitioner

" My role in the company is to guide and manage the chip design team that is doing a lot of interesting work in developing new chips that will enable new and better products in the computer peripheral industry,"

Rahul Hakhoo
Head, Computer Peripheral Divisions
ST Microelectronics

What are the required skill sets?

- Basic requirement is a degree in Electronics Communication Engineering. During interviews, firms look for a good sense of engineering with a sound grounding in mathematics, physics and electronics. Knowledge of different languages is not a must.
- A number of courses have come up in design which teach the tools (VHDL, EDA, etc.) and taking them would be a bonus.
- Constant improvement of skills and knowledge is a must as the design tools and processes keep evolving.

What are the future prospects?

Compared to other engineering fields, for example the mechanical industry, the chip industry is more cyclical in nature. If one is making a career decision,